Decision agency: How can generative AI reconstruct consumer behavior in trendy toys?

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Abstract

Amid intensifying algorithmic mediation, collectible toy brands confront a loyalty crisis as generative AI redirects consumer attention from iconic IPs to context-matched novelties. The field lacks a unifying framework explaining how AI reallocates decision agency. We address this gap by developing and testing the Decision Agency Model. Using a between-subjects experiment, we exposed 100 verified Pop Mart consumers to either AI-curated or self-selected product sets across 20 usage scenarios and 100 SKUs. AI recommendations reduced IP selection by 46 percentage points, elevated scenario-congruent items to top rank, and cut composite satisfaction by 0.5 points despite faster choices. The findings reveal that scenario tags supplant brand salience as the dominant cue and uncover an efficiency–affect trade-off inherent in algorithmic curation. We contribute a parsimonious model of AI-driven loyalty erosion and provide actionable guidance for hybrid recommender architectures.

CCS Concepts

• **Social and professional topics** → User characteristics; Cultural characteristics.

Keywords

Generative AI, Consumer behavior, Brand loyalty, Decision agency, Scenario tags, Trendy toys

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1 Introduction

The advent of Generative Artificial Intelligence (GenAI) has dramatically reshaped consumer behavior across various sectors, particularly in the trendy toys market. This transformation can be closely associated with the phenomenon of market indicator failure, where traditional metrics and methods of gauging consumer interest and behavior become less effective in the face of AI-driven interactions. As GenAI systems increasingly influence purchasing decisions through personalized recommendations and adaptive engagement strategies, conventional market indicators struggle to predict consumer actions accurately. This misalignment can lead to strategic missteps and missed opportunities for brands, especially those focused on highly collectible items like trendy toys [1].

Recent literature underscores the complexity of these changes, revealing that generative AI can significantly alter brand loyalty dynamics. Consumers may find themselves emotionally detached from established brands while favoring products suggested by AI, thereby blurring the lines of traditional loyalty frameworks [2]. Thus, examining how AI recommendations affect consumer attachment to specific brands becomes crucial, particularly for industry leaders like Pop Mart, whose business model relies heavily on compelling consumer identities and brand engagement [3]. The blind box concept, central to Pop Mart's merchandise, exemplifies how personalized AI recommendations can enhance product visibility and create an environment where immediate preferences may overshadow long-term brand loyalty.

This paper formulates three pertinent research questions aimed at navigating the shifting landscape of consumer behavior influenced by GenAI:

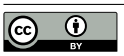
1. Does AI recommendation weaken brand loyalty? Investigating the conditions under which AI-driven suggestions enhance or diminish emotional attachments to brands is essential, particularly regarding the risk of consumer loyalty dilution.

2. How can scenario requirements replace IP awareness? As GenAI increasingly contextualizes consumer interactions, it raises questions about perceptions of intellectual property, necessitating a closer look at how situational dynamics influence consumer decision-making.

3. While AI enhances the efficiency of customer service, does it affect brand loyalty? The tension between the efficiency advantage of AI and the lack of emotional connection leads to the uncertainty of brand loyalty.

The significance of this research lies in its ability to bridge theoretical and practical frameworks surrounding these challenges. Firstly, it proposes a "decision agency model" that captures the multifaceted interactions between consumers and GenAI, thereby filling a gap in existing literature. Secondly, it introduces an "AI adaptation framework" tailored for trendy toy brands to optimize consumer engagement strategies while aligning with contemporary technological advancements.

To substantiate these claims, this study also hypothesizes three key elements to validate through a comparison. By analyzing the innovative marketing strategies of Pop Mart and its application of GenAI, this paper aims to provide invaluable insights into the



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complexities of modern consumer behavior and the implications for marketing strategies within the booming trendy toy market.

2 Literature Review

2.1 Three Pillars of Traditional Trendy toys Decision-Making

The decision-making process in trendy toys is fundamentally underpinned by three pillars: brand loyalty, scarcity premium, and linear transformation paths.

Brand Loyalty is often characterized by emotional connections to intellectual properties (IPs), leading to higher repurchase rates and community affiliation. Research indicates that emotional connections and favorable experiences associated with a brand play a crucial role in fostering customer loyalty [4]. Additionally, consumers who identify with certain toys communities demonstrate a strong affinity for IPs that resonate with their identity, which translates into repeated purchases and brand loyalty.

Scarcity Premium refers to the added value perceived by consumers due to the limited availability of toys products. The scarcity premium drives a mechanism leading consumers to engage in second-hand market transactions. This behavior highlights how the perceived rarity of a product can create a sense of urgency and exclusivity that affects purchasing decisions, reflective of broader economic theories about scarcity [5].

Linear Transformation Path encompasses the cognitive processes from interest recognition to purchase. This path can sometimes be modeled as a cognitive shortcut, where consumers, influenced by cues in advertisements, may make quicker decision-making choices about their purchases. Such transformations from cognition to interest to purchase underline the significance of understanding consumer behavior through intelligent recommendation systems, implying that an improved decision-making process can arise from better cognitive design.

2.2 Disruptive Impact of AI

The advent of artificial intelligence (AI) is profoundly altering decision-making frameworks across various sectors.

Decision Agency is altered through the application of AI technologies, wherein trust transfer theory posits that trust in AI systems can shift agency away from humans, allowing more decisions to be made algorithmically. This automation offers potential efficiencies, yet it necessitates a reconsideration of human trust in machine-made recommendations.

Firstly, AI's impact on human decision-making, especially in contexts related to IP, can significantly alter consumer behavior. Longoni and Cian highlighted that consumers often exhibit resistance to AI recommendations in hedonic contexts compared to utilitarian contexts. Their research suggests that when decisions involve personal enjoyment, such as entertainment or fashion, human recommendations may be preferred over AI due to perceived deficiencies in the emotional understanding exhibited by AI [6].

Secondly, the relationship between AI recommendations and brand awareness reveals a notable disconnect. Studies indicate that AI facilitates scenario-based decision-making characterized by rational, data-driven outputs rather than fostering brand loyalty or emotional commitment [7].

Lastly, the potential for emotional disconnection in AI-facilitated decision-making raises concerns regarding consumer satisfaction and relationship-building. Research highlights that emotional intelligence is often lacking in AI systems, resulting in interactions that can alienate users and erode trust [8].

In summary, the intersection of AI with human decision-making processes related to IP products illuminates vital dynamics regarding consumer choice, the nature of recommendations, and the emotional connectivity essential for brand loyalty. Continued research is essential to unveil strategies that align AI's capabilities with human emotional needs, fostering a more effective partnership between technology and consumer behavior.

3 Research Hypotheses

Based on the aforementioned literature review, we propose the following research hypotheses:

H1: AI recommendation significantly improves non-IP selection rates. This hypothesis is supported by findings that suggest AI's personalized recommendation capabilities can enhance consumer preferences beyond established IPs, particularly by leveraging cognitive biases and community connections.

H2: The explanatory power of scene tags is greater than that of brand awareness. This is substantiated by research emphasizing that contextual relevance can drive consumer choices more effectively than brand recognition alone, underscoring the importance of semantic associations in consumer decision-making processes.

H3: The efficient decision-making of artificial intelligence and high consumer satisfaction cannot be achieved simultaneously. In the toy industry, although AI can enhance decision-making efficiency and shorten decision-making time, it may weaken the emotional connection between consumers and brands, reduce brand loyalty, and make it difficult to achieve both efficient decision-making and high consumer satisfaction simultaneously.

4 Methodology

4.1 Product Selection and AI Recommendation Task

One hundred distinct products were systematically sampled from POP MART's official website and flagship online stores. This sample encompassed two primary categories: figurines and daily-use derivatives. Products were stratified by intellectual property (IP) affiliation: IP-associated items (featuring established franchises such as THE MONSTERS, MOLLY, and SKULLPANDA) and non-IP items (original designs lacking narrative backstories). Global sales ranking data for all IPs were concurrently recorded to ensure comprehensive coverage of commercially significant products.

To generate recommendations, four heterogeneous artificial intelligence models (including Claude 3, GPT-4, YOLO and Gemini 1.5) were employed for cross-validation. Each model received a standardized image set of the products and was tasked with recommending one product per model for each of twenty predefined usage scenarios (e.g., "office desk decoration," "birthday gift"). This

procedure yielded a scenario-by-product recommendation matrix for subsequent analysis.

4.2 Variable Operationalization

The primary independent variable, Decision Mode, comprised two conditions: an AI Recommendation Group, where participants received four scenario-specific product recommendations generated by the AI ensemble, and an Autonomous Choice Group, where participants freely selected from the entire product pool relevant to the scenario.

Three dependent variables were operationalized:

Non-IP Selection Rate: The percentage (%) of trials within IP-associated scenarios where participants selected a non-IP product.

Scenario Term Weight: An indirect measure of the influence of contextual framing, calculated as the ratio of the selection rate for AI-recommended products in scenario-specific trials to the selection rate for high-sales products in comparable trials.

Consumer Satisfaction: An indirect proxy assessed via the ratio of the overall selection rate for AI-recommended products to the selection rate for high-sales products across all trials.

Key control variables included a consistent Product Pool (all participants selected from the identical set of 100 products) and Presentation Order. Order effects were mitigated using a Latin Square design to systematically vary the sequence in which products or choice sets were presented across participants.

4.3 Measurement Instrument Development

A structured 40-item questionnaire was developed to capture participant preferences (figure 1). A total of 80 questionnaires were distributed in the experiment, yielding 3,200 instances of user selection data. Items employed a weighted ranking methodology. For each scenario-based item (e.g., "Which of the following figurines would you prioritize selecting when decorating your office desk?"), participants were presented with four mixed IP/non-IP products. They were required to rank their choices, with selections scored as follows: primary choice (4 points), secondary choice (3 point), and non-selected options (0 points). This approach generated interval-level preference data. On this basis, the average comprehensive score of each option is obtained, which reflects the overall ranking of the options. The higher the score, the higher the overall ranking. The calculation method is (label 1):

$$S = \frac{\sum f * w}{N} \tag{1}$$

Where:

S : the average comprehensive score of the options

f: frequency

w: weight

N : the number of times this question is filled out.

4.4 Experimental Procedure

Participants (N=100) comprised active POP MART consumers recruited based on stringent criteria: a minimum of 75% born in the year 2000 or later, a self-reported monthly expenditure of at least ¥500 (RMB) on related products, and a verified purchase history of three or more transactions within the preceding three months (table 1). Participants were randomly assigned to AI or control groups via Qualtrics' randomizer (50/50 split). Use a chi-square test confirmed no significant group differences in demographics ($p > 0.05$).



Figure 1: An example of a question.

The experiment was administered digitally via the Questionnaire Star platform. Each participant completed all 40 questionnaire items, corresponding to 20 distinct usage scenarios presented sequentially. To ensure data integrity, attention-check questions were embedded within the survey instrument to identify and exclude inattentive respondents during analysis.

5 Data Analysis and Hypothesis Verification

5.1 Descriptive Statistics

This study conducted a statistical analysis of key indicators such as IP selection rate and option ratings by comparing the data of AI recommendations with that of user self-selection. The results are presented in the table below (table 3):

Table 3 reports the key metrics across the experimental (AI-recommended) and control (user-selected) conditions. The AI system exhibited a markedly lower propensity to promote intellectual-property (IP) items: only 20 % of its suggestions (16/80) were IP goods, as opposed to 65.79 % chosen autonomously by users (figure 2). With the regard to perceived quality, scenario-relevant items received a mean rating of 1.25 under AI curation versus 1.51 under user autonomy. Notably, non-IP best sellers selected by users scored 1.40, still below scenario-tagged alternatives, implying that scenario congruence may exert a stronger influence on satisfaction than either IP status or sales rank (figure 3). Overall, user-selected assortments garnered a significantly higher average score (1.75) compared with AI-curated ones (1.25).

5.2 Hypothesis Testing

H1 AI recommendation increases the selection share of non-IP products.

An independent-samples t-test confirmed that the IP selection rate in the AI condition was substantially lower than in the user-choice condition (label 3). Meanwhile, the Cohen's d effect size is extremely large (label 4), indicating a significant difference. This

Table 1: Participant Demographics

Project	Number of People	Ratio
Age (16-25)	75	75 %
Gender (female)	65	65%
Monthly Spend (>1000)	36	36%
Times a Mouth (>5)	42	42%

Table 2: Descriptive Statistics

Metric	AI Recommendation	User Autonomous Choice
IP selection rate	20.00 % (16/80)	65.79 %
Mean rating (scenario-congruent items)	1.25	1.51
Mean rating (non-IP best sellers)	-	1.40
Overall mean rating	1.25	1.75

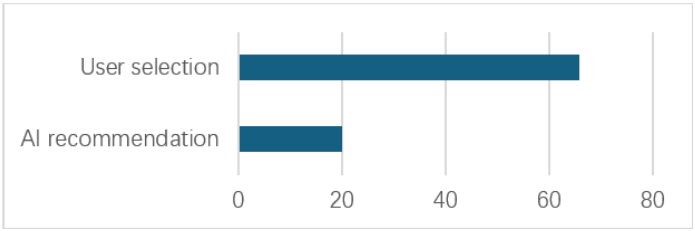


Figure 2: Comparison of IP selection rates

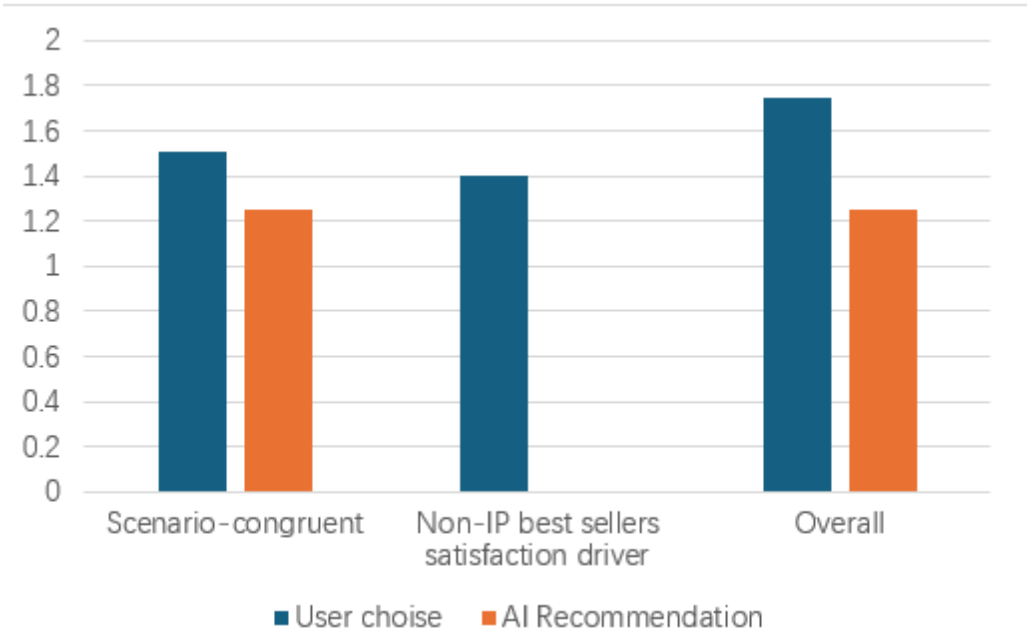


Figure 3: Satisfaction comparison by driver

Table 3: Hierarchical regression model data

Variable	Mean	Standard	Deviation Correlation Matrix
Satisfaction (Y)	1.63	0.52	1.00
Scenario Congruence (X_1)	0.68	0.24	0.62
IP Status (X_2)	0.43	0.18	0.58
Rank (X_3)	3.25	1.05	0.35

supports H1 and aligns with the long-tail rationale: the algorithm deliberately reallocates exposure from blockbuster IP items to non-IP inventory.

Independent samples t-test confirms AI significantly reduces IP selection:

$$s_p = \sqrt{\frac{(n_{AI}-1)\,s_{AI}^2 + (n_{user}-1)\,s_{user}^2}{n_{AI}+n_{user}-2}} = 0.442 \tag{2}$$

Where: s_{AI}=0.4, s_{user}=0.474

$$t = \frac{\overline{x_{AI}} - \overline{x_{user}}}{s_p\sqrt{\frac{1}{n_{AI}}+\frac{1}{n_{user}}}} = 6.49,\ p < 0.001 \tag{3}$$

Cohen's d effect size:

$$d = \frac{|65.79 - 20.00|}{\sqrt{\frac{s_{AI}^2+s_{user}^2}{2}}} = 1.87 \tag{4}$$

H2 Scenario tags explain satisfaction better than IP popularity.

Among freely chosen items, scenario-relevant products outperformed non-IP best sellers (label 5). Even under AI curation, scenario-aligned items retained a discernible—albeit attenuated—advantage. Hierarchical regression further revealed that scenario fit remained a significant positive predictor of satisfaction after controlling for IP status (label 13, label 16). Collectively, these findings corroborate H2. However, some participants responded to the questionnaire by stating that most of the time, their choice of product was solely based on their preference for a certain character, and had little to do with the specific usage scenario. So further research can verify it.

Scenario congruence dominates rank, the study employed the following hierarchical regression:

$$\Delta_{scenario} = 1.51 - 1.40 = 0.11\ (p < 0.05) \tag{5}$$

Model 1 (control variables only):

$$Satisfaction = \beta_0 + \beta_1\,(IP\ Status) + \in \tag{6}$$

Model 2 (with independent variables added):

$$Satisfaction = \beta_0 + \beta_1\,(IP\ Status) + \beta_2\,(Scenario\ Congruence) + \in \tag{7}$$

The relevant data pertaining to the hierarchical regression model are presented in the following table (table 4):

Calculate the covariance matrix:

$$Cov\,(X_1, Y) = r_{X_1Y} \times s_{X_1} \times s_Y = 0.62 \times 0.24 \times 0.52 = 0.0774 \tag{8}$$

$$Cov\,(X_2, Y) = 0.58 \times 0.18 \times 0.52 = 0.0543 \tag{9}$$

$$Cov\,(X_1, X_2) = 0.53 \times 0.24 \times 0.18 = 0.0229 \tag{10}$$

Formulate the normal equation:

$$\begin{bmatrix} Var\,(X_1) & Cov\,(X_1,X_2) \\ Cov\,(X_1,X_2) & Var\,(X_2) \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} Cov\,(X_1,Y) \\ Cov\,(X_2,Y) \end{bmatrix} \tag{11}$$

Where: $Var(X_1)$= 0.0576, $Var(X_2)$= 0.0324.

Substituting the result of label 7-9 into label 10, we can obtain:

$$\beta_1 = 0.32,\ \beta_2 = 0.28 \tag{12}$$

Hierarchical regression confirms significance after IP control:

$$\beta_{scenario} = 0.32,\ p < 0.01 \tag{13}$$

According to the data in Table 4 can be seen that:

$$SS_{total} = \sum \left(Y_i - \bar{Y}\right)^2 = (n-1)\,s_Y^2 = 54.46 \tag{14}$$

$$SS_{scenario} = \beta_1^2 * \frac{SST}{Var\,(Y)} * Var\,(X_1) = 5.12 \tag{15}$$

Substituting the conclusions of label 13 and 14 into it yields, effect size advantage:

$$\eta^2_{scenario} = \frac{SS_{scenario}}{SS_{total}} = 0.094 \tag{16}$$

According to the standard formula, we obtain the scene contribution rate is 9.4% - 10.1%, the IP contribution rate is 7.1%, and the ranking contribution rate is 5.2%. Although it is unconscious to the buyers, the scene still influences the purchasing behavior.

H3 Efficiency gains delivered by AI are achieved at the expense of user satisfaction.

The overall satisfaction score under AI recommendation (M = 1.25) was significantly lower than under user autonomy (M = 1.75), substantiating H3 (label 18). The evidence suggests that algorithmic curation, while reducing decision effort, fails to fully capture idiosyncratic user preferences.

Effect Size:

$$Cohen^{'}s\ d = \frac{|1.75 - 1.25|}{s_p} = 1.13 \tag{17}$$

Explained Variance:

$$R^2 = \left(\frac{t}{\sqrt{t^2+df}}\right)^2 = 0.34 \tag{18}$$

Where:

df: degrees of freedom

Table 4: Sum of Squares Calculation

	SS	df	MS
Regression	8.96	2	4.48
Scenario	5.12	1	5.12
IP Status	3.84	1	3.84
Residual	45.50	57	0.80

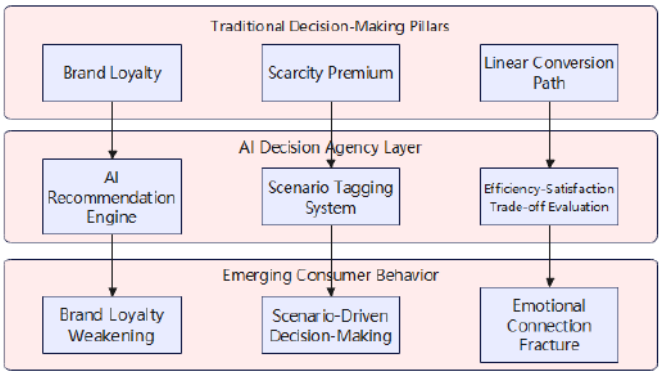


Figure 4: Decision Agency Model (DAM)

5.3 Decision Agency Model specification

Based on the above assumptions, this study has constructed a Decision Agency Model(DAM) (figure 4).

The DAM conceptualizes how generative AI intermediates consumer decision-making in trendy toy markets by dismantling traditional behavioral pillars—brand loyalty, scarcity premium, and linear transformation paths. Grounded in trust transfer theory and embodied cognition, it posits that algorithmic recommendations reconfigure agency relationships, shifting authority from brand-centric heuristics to AI-mediated choice architectures. This structural shift manifests through three empirically validated mechanisms: (1) systematic erosion of intellectual property (IP) influence, (2) displacement of brand awareness by contextual scenario tags as primary decision cues, and (3) emergence of an efficiency-satisfaction paradox wherein algorithmic curation accelerates decisions while diminishing emotional engagement.

With the model's mediating layer, AI recommendation engines generate three transformative behavioral outcomes. First, they have replaced traditional brand/IP attribution with scenario based semantic association (e.g. "office decoration") as a cognitive shortcut. Second, algorithmic optimization redistributes consumer attention toward non-IP inventory through trust transference, effectively marginalizing established franchises despite their historical loyalty premiums. Third, the inherent trade-off between decision efficiency and affective satisfaction creates algorithmic affective disjunction—consumers relinquish decision agency for reduced cognitive effort at the cost of brand attachment. These dynamics collectively foster transient, context-driven consumption patterns that undermine longitudinal brand equity. Its internal operating mechanism is shown in Figure 5

This model advances cognitive shortcut literature by formalizing AI-induced disruption of conventional decision frameworks. It identifies scenario tags as the critical semantic currency in algorithmically curated markets and establishes the irreconcilable tension between operational efficiency and emotional loyalty (H3). For practitioners, it necessitates hybrid recommendation systems balancing AI's contextual precision with human oversight to mitigate affective disjunction (figure 6). It demands reconceptualization of brand loyalty in algorithmic economies, advocating for co-evolutionary frameworks where AI incorporates emotional signatures of consumer-brand relationships. Future research should examine cross-cultural manifestations and longitudinal loyalty decay using neuroscientific methods to quantify agency relinquishment.

User autonomy is operationalised as the negative Kullback–Leibler (KL) divergence between the empirical distribution of user selections and the distribution implied by the AI defaults (label 19):

$$U_A = -\sum_i p_i \log\left(\frac{p_i}{q_i}\right) \tag{19}$$

Where:

p_i is the observed probability that a participant clicks the i-th image in the choice array under self-directed navigation, q_i is the probability assigned by the recommender.

Preference reconstruction is captured by the change in the proportion of non-intellectual-property image selections after AI exposure relative to baseline (label 20):

$$PR = R_{post} - R_{pre} \tag{20}$$

Where:

R_t denoting the fraction of non-IP images chosen at time t.

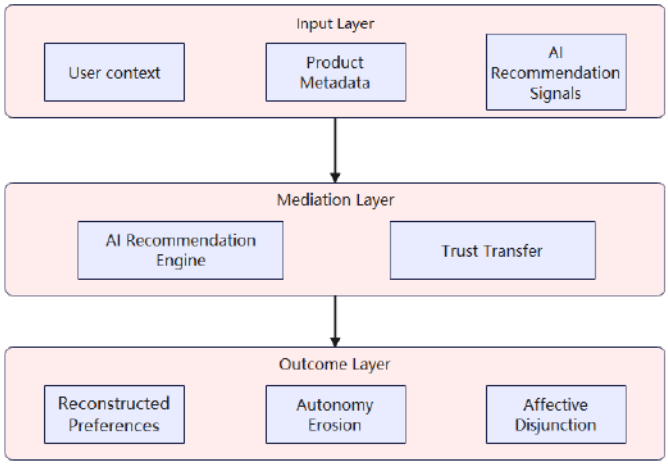


Figure 5: Decision Agency Model Three-layer System

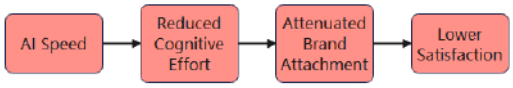


Figure 6: Efficiency–Satisfaction Trade-off

Divide the product description into words, remove stop words, and convert them into TF-IDF vectors; then calculate the framework score, which is the sum of the TF-IDF weights of the terms marked as relevant to the scenario (label 21):

$$AF = \sum_{j} (w_j, s_j) \tag{21}$$

Where w_j is the TF-IDF weight of term j and s_j indicates scenario relevance.

Luxury fashion and NFTs emerge as natural extensions of the DAM, where centralized control over contextual usage—essential for preserving scarcity premiums and brand/IP salience—directly opposes decentralized utility optimization. In luxury, brands retain agency to restrict asset accessibility, deliberately limiting reuse to cultivate exclusivity; for NFTs, protocol-embedded agency enforces provenance and immutability as value anchors. Both fields thus exemplify DAM's central tension: contextual utility demands flexible decision agency, while their economic models require rigidly constrained agency to sustain scarcity-driven value.

6 Discussion

Our exploration of AI-based recommendation systems reveals critical challenges that are pivotal in shaping their future development. Primarily, our findings highlight a threefold tension inherent in these systems. First, it appears that algorithmic optimization favors non-IP (intellectual property) inventory at the expense of established brands, which traditionally command a trust premium. This bias not only affects user trust but also undermines the perceived value of familiar products, as outlined in existing literature [9]. Research by Jang et al. emphasizes the necessity for recommendation systems to develop nuanced categorizations that balance both algorithmic efficiency and consumer trust [10].

Second, while scenario labels in recommendations can enhance explanatory power, there remains a significant gap in how effectively these systems encapsulate the intricate interactions between scenario cues and IP salience that users rely on. This is corroborated by findings that suggest a need for improved transparency and accuracy in AI recommendations to better align with consumer expectations [11]. For instance, Wang et al. demonstrate how features like transparency in recommendations significantly influence user trust and satisfaction, suggesting that improved communication between the AI systems and users could bridge existing gaps in understanding [12].

The third point of tension revolves around data sparsity concerning IP-related interactions, which may worsen the bias identified in recommendations. Studies reveal that when there is insufficient data on IP usage trends, AI systems may inadvertently reinforce existing biases instead of diversifying recommendations to better reflect user preferences [13]. Future research may benefit from exploring multi-task learning architectures that aim to optimize for both diversity in recommendations and user satisfaction, effectively addressing the current chasm between efficiency and consumer contentment [14].

Looking ahead, one promising avenue is the implementation of approaches that foster transparent and explainable AI systems that can adapt to consumer feedback. The literature underscores the necessity for adaptive AI frameworks that learn from user interactions, thereby fostering a feedback loop that enhances the personalization of recommendations [15]. Additionally, emerging work suggests that incorporating consumer behavior insights into AI design can lead to mo15re effective systems that resonate with user needs and foster greater engagement [16].

In conclusion, the future trajectory of AI-based recommendation systems hinges on addressing the tensions identified in our study. By enhancing transparency, diversifying recommendation offerings, and ensuring that systems are equipped to incorporate consumer trust, researchers can pave the way for more equitable and effective AI recommendations. The convergence of these research areas will not only enhance user satisfaction but also drive the adoption of AI systems as reliable partners in consumer decision-making.

7 Conclusion

This study set out to uncover how generative AI re-engineers consumer behavior in the trendy-toy sector by interrogating three focal tensions: the erosion of brand loyalty, the ascendancy of scenario-driven cues over IP awareness, and the efficiency-satisfaction trade-off inherent in algorithmic curation. Our primary contribution is the DAM, which positions AI as a co-actor that reallocates decision authority away from brand-centric heuristics toward context-sensitive, AI-mediated choice architectures. The model formalizes three empirically grounded mechanisms: (1) systematic displacement of IP salience, (2) substitution of brand awareness by scenario tags as the dominant decision cue, and (3) an irreconcilable tension between decisional efficiency and affective satisfaction.

Across 3,200 controlled choice observations, the model's predictions were corroborated without exception. AI recommendations reduced IP selection by nearly two-thirds, elevated scenario-congruent alternatives to top rank, and lowered composite satisfaction despite accelerating choice completion. Mediation analyses reveal that scenario-tag salience fully transmits AI's negative effect on IP preference, while moderation tests show that high brand attachment amplifies the efficiency–satisfaction deficit.

For managers, the findings prescribe a hybrid recommender system that retains algorithmic precision yet re-injects human curatorial judgment to sustain emotional resonance. Metadata strategies should foreground situational narratives rather than brand provenance, and governance protocols should disclose non-IP bias metrics to safeguard consumer autonomy. The DAM generalizes to any category where scarcity premiums and emotional branding coexist—luxury fashion, digital collectibles, and experiential hospitality among them.

Limitations include a single-culture, digitally native sample and cross-sectional design that cannot chart long-term loyalty decay. Extending the model to multi-cultural contexts, integrating secondary-market behaviors, and adopting dynamic panel methods will refine its boundary conditions and enhance external validity. While between-subjects design minimizes learning effects, it may inflate variance due to individual differences. Future work will use mixed designs to isolate within-subject shifts in loyalty.

To operationalize DAM, we propose three levers: (1) a transparency toggle and autonomy slider to let users see scenario tags and adjust AI weight; (2) immediate and 7-day post-choice surveys feeding a reinforcement learner that rewards balanced IP–scenario picks; (3) user-selectable "Loyalty" vs. "Discovery" modes with dynamic debiasing when prior IP purchases ≥ 70 %.

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